MTH 2215- Midterm Exam

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General instructions:

- Shows all your work to receive full credit.
- No discussion is allowed in the exam room.

PROBLEM	POINTS	SCORE
1	40	40
2	40	40
3	40	40
4	40	40
5	40	40
6	30	30
7	40	40
8	30	30
TOTAL	300	300

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1. (40 points) i) Find the truth table of the proposition

$$\neg(p \wedge q) \vee (\neg q \vee r)$$

ii) Determine the value of each of the following proposition given:

$P(x, y) : (x < y) \rightarrow (x^2 < y^2)$, x, y are real numbers

a) $\forall x \forall y P(x, y)$ b) $\exists y \forall x P(x, y)$

i) The truth table of the proposition. $\neg(p \wedge q) \vee (\neg q \vee r)$.

p	q	r	$p \wedge q$	$\neg(p \wedge q)$		$\neg q \vee r$	Result
T	T	T	T	F		T	T
T	T	F	T	F		F	F
T	F	T	F	T		T	T
T	F	F	F	T		T	T
F	T	T	F	T		T	T
F	T	F	F	T		F	T
F	F	T	F	T		T	T
F	F	F	F	T		T	T

ii, $P(x, y) : (x < y) \rightarrow (x^2 < y^2)$, $x, y \in R$

a, $\forall x \forall y P(x; y)$.

For every x, y, if $x < y$, then $x^2 < y^2$.

Take a counterexample: $x = -2$; $y = 1$.

$\rightarrow -2 < 1$ (true).

But $(-2)^2 = 4$; $1^2 = 1$.

$\Rightarrow 4 < 1$ (false) ✓

Hence $P(-2; 1)$ is false.

Therefore: $\forall x \forall y P(x; y)$ is false.

b, $\exists y \forall x P(x; y)$

There exists a real number y such that for every real x, $(x < y) \rightarrow (x^2 < y^2)$.

We show this is impossible.

For any real y, choose:

- if $y \geq 0$, let $x = y - 1$.
- if $y < 0$, let $x = y - 1$.

In both case: $x < y$ and $x^2 < y^2$.

$\Rightarrow$ The implication fails.

Thus no such y exists.

Therefore: $\exists y \forall x P(x, y)$ is false. ✓

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2. (40 points) i) Prove the following set identity:

$$(A-B)\cup(B-A) = (A\cup B)-(A\cap B)$$

ii) Is the set identity $(A-B)\cup(B-C) = (A-C)$ true? If it is true give a proof, otherwise give a counterexample to show that it is false.

i). $(A-B)\cup(B-A) = (A\cup B)-(A\cap B)$

$A-B = \{x \mid x \in A \text{ and } x \notin B\}$.

$B-A = \{x \mid x \in B \text{ and } x \notin A\}$.

So, $(A-B)\cup(B-A)$ contains all elements that belong to exactly one of the two sets A or B.

The right-hand side: $(A\cup B)-(A\cap B)$.

$\Rightarrow$ $\begin{cases} x \in A\cup B : x \text{ is in at least one of } A, B. \\ x \notin A\cap B : x \text{ is not in both.} \end{cases}$

$\Rightarrow$ x belongs to exactly one of A or B.

Thus both sides describe the same sets:

$(A-B)\cup(B-A) = (A\cup B)-(A\cap B)$.

ii), Answer: false:

Let $A = \{1\}$; $B = \{2\}$; $C = \{3\}$.

Left-hand side: $(A-B)\cup(B-C) = (\{1\}-\{2\})\cup(\{2\}-\{3\}) = \{1\}\cup\{2\} = \{1,2\}$.

Right-hand side: $(A-C) = \{1\}-\{3\} = \{1\}$.

Since $\{1,2\} \neq \{1\}$, the identity is false. ✓

3. (40 points) i) For each of the following predicates, find all the values in the domain $D = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$ such that the predicate is true:

a) $Q(n): (n^2 < 15) \rightarrow (n \text{ is an even number})$

b) $S(n): (2n > 10) \vee (n \mid 8)$

ii) Prove that $n = \left\lfloor \frac{n}{2} \right\rfloor + \left\lceil \frac{n}{2} \right\rceil$ for any positive integer n.

i/ a, $Q(n)$: $(n^2 < 15) \rightarrow n$ is an even number.

- Test n where $n^2 < 15$ is true. These are $n \in \{1; 2; 3\}$.
  + $n = 1$: $(1 < 15) \rightarrow (1 \text{ is even}) \equiv T \rightarrow F \equiv$ False.
  + $n = 2$: $(4 < 15) \rightarrow (2 \text{ is even}) \equiv T \rightarrow T \equiv$ True.
  + $n = 3$: $(9 < 15) \rightarrow (3 \text{ is even}) \equiv T \rightarrow F \equiv$ False.
- Test n where $n^2 < 15$ is false: These are $n \in \{4, 5, 6, 7, 8, 9\}$.
  + Since the hypothesis is false, the implication is true.

=) Values for which $Q(n)$ is true: $\{2; 4, 5, 6, 7, 8, 9\}$ ✓

b, $S(n)$: $(2n > 10) \vee (n \mid 8)$.

- Check $2n > 10$. This is true for $n > 5$, so $n \in \{6, 7, 8, 9\}$.
- Check $n \mid 8$. The divisors of 8 in the domain are $n \in \{1, 2, 4, 8\}$.
- Union of sets: $\{6, 7, 8, 9\} \cup \{1, 2, 4, 8\} = \{1, 2, 4, 6, 7, 8, 9\}$.

=) Values for which $S(n)$ is true: $\{1, 2, 4, 6, 7, 8, 9\}$ ✓

ii, $n = \left\lfloor \frac{n}{2} \right\rfloor + \left\lceil \frac{n}{2} \right\rceil$

Case 1: n is even.

Let $n = 2k$ for some integer k.

+ $\left\lfloor \frac{2k}{2} \right\rfloor = \lfloor k \rfloor = k$.
+ $\left\lceil \frac{2k}{2} \right\rceil = \lceil k \rceil = k$.
+ Sum: $k + k = 2k = n$ (correct).

Case 2: n is odd.

Let $n = 2k + 1$ for some integer k.

+ $\left\lfloor \frac{2k+1}{2} \right\rfloor = \lfloor k + 0{,}5 \rfloor = k$.
+ $\left\lceil \frac{2k+1}{2} \right\rceil = \lceil k + 0{,}5 \rceil = k + 1$
+ Sum: $k + (k+1) = 2k + 1 = n$ (correct).

=) Since the identity holds for both even and odd integers, the proof is complete

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4. (40 points) Which of the following functions are injective? surjective?bijective? Explain your answers.

- $f:\mathbb{N}\to\mathbb{N}, f(x)=\begin{cases}\frac{n+1}{2}, & n\equiv 1 \bmod 2\\ \frac{n}{2}, & n\equiv 0 \bmod 2\end{cases}$ where $\mathbb{N}=\{0,1,2,3,\ldots\}$
- $g:\mathbb{R}\to\mathbb{Z}, g(x)=\lfloor x+1\rfloor$
- $h:\mathbb{Z}^+\to\mathbb{Z}, h(x)=x^2-1$ where $\mathbb{Z}^+=\{1,2,3,\ldots\}$

(*) $f: \mathbb{N}\to\mathbb{N}$, $f(x)=\begin{cases}\frac{n+1}{2}, & n\equiv 1 \bmod 2.\\ \frac{n}{2}, & n\equiv 0 \bmod 2.\end{cases}$ where $\mathbb{N}=\{0;1;2;3;\ldots\}$.

- Injective : No. ✓
  + Example: $f(0)=\frac{0}{2}=0$ and $f(-1)$ isn't in the domain, but check $f(1)$ and $f(2)$.
  + $f(1)=\frac{1+1}{2}=1$.
  + $f(2)=\frac{2}{2}=1$.
  + Since $f(1)=f(2)$ but $1\neq 2 \Rightarrow$ it's not injective.
- Surjective : Yes. ✓
  + For any $y\in\mathbb{N}$, we can choose $n=2y$. Since n is even, $f(2y)=\frac{2y}{2}=y$. Every element in the codomain is hit.
- Bijective : No (it's not injective).

(*) $g: \mathbb{R}\to\mathbb{Z}$, $g(x)=\lfloor x+1\rfloor$

- Injective : No. ✓
  + The floor function maps many real numbers to the same integers
  + Example: $g(0,1)=\lfloor 1,1\rfloor=1$ and $g(0,2)=\lfloor 1,2\rfloor=1$.
- Surjective : Yes.
  + For any integer $z\in\mathbb{Z}$, let $x=z-1$. Then $g(z-1)=\lfloor(z-1)+1\rfloor=\lfloor z\rfloor=z$.
- Bijective: No (it's not injective). ✓

(+) $h: Z^{+} \to Z$, $h(x) = x^2 - 1$ where $Z^{+} = \{1; 2; 3; \dots\}$.

→ Injective: Yes:

+ Suppose $h(a) = h(b) \Rightarrow a^2 - 1 = b^2 - 1 = a^2 - b^2$.

+ Since the domain is restricted to positive integers ($a, b > 0$) then $a = b$.

– Surjective: No.

+ The codomain is all integers Z, but $x^2 - 1$ cannot produce all integers.

+ Example: There is no $x \in Z^{+}$ such that $x^2 - 1 = 1$ (because $x^2 = 2$ has no integer solution) or $x^2 - 1 = -5$.

+ Bijective: No (it's not surjective).

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5. (40 points) What is the least integer n such that f(x) is $O(x^n)$ for each of the following functions? Explain in detail.

a) $f(x) = x^{3/2} + x^2 + 5$

b) $f(x) = 2x^3 + x^4 \ln x - 2x^2$

a, $g(x) = x^{5/2} + x^2 + 5$

+ The dominant term: As x becomes very large, x^2 grows faster than $x^{1.5}$ and 5.

+ Big-O bound: Since x^2 is the highest power, $g(x)$ is $O(x^2)$.

+ Conclusion: The least integer n is 2. ✓

b, $g(x) = 2x^5 + x^4 \ln x - 2x^2$.

+ We know that every $\epsilon > 0$, $\ln x$ grows slower than x^ϵ

+ Therefore, $x^4 \ln x$ grows faster than x^4, but slower than $x^{4+\epsilon}$

+ If we pick $n = 4$, then $\frac{x^4 \ln x}{x^4} = \ln x$, which approaches ∞ as $x \to \infty$.
Thus, $g(x)$ is not $O(x^4)$.

+ If we pick $n = 5$, then $\frac{x^4 \ln x}{x^5} = \frac{\ln x}{x}$, which approaches 0 as $x \to \infty$.

Thus $g(x)$ is $O(x^5)$.

+ Conclusion: The least integer n is 5. ✓

6. (30 points) Let x, y be positive real numbers such that $x^2 - y^2 = 1$. Prove that x or y (or both) is not an integer.

We have: $x^2 - y^2 = (x-y)(x+y) = 1$.

Assume, for contradiction, that both x and y are integers.

Since, $x, y \in \mathbb{Z}$, it follows that x^2 and y^2 are integers.

Hence: $x^2 y^2 = 1$. implies: $x^2 = y^2 + 1$.

Thus two perfect squares differ by 1.

Now let $y = n$, where $n \in \mathbb{Z}^+$. Then. $x^2 = n^2 + 1$.

Since: $(n+1)^2 = n^2 + 2n + 1 > n^2 + 1$ (because $n \geq 1$).

We obtain: $n^2 < n^2 + 1 < (n+1)^2$

So $x^2 = n^2 + 1$ lies strictly between two consecutive perfect squares n^2 and $(n+1)^2$, which is impossible. Therefore, our assumption that both x and y are integers is false.

Hence at least on of x or y is not an integer.

=) Therefore, x or y (or both) is not an integer. ✓

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7. (40 points) i) Show the steps of using Euclidean algorithm to find gcd(1411,2703)
ii) Find the binary representation of $(8FB)_{16}$
iii) Find the octal representation of $(10010110101)_2$

i, gcd(1411, 2703).

Apply the Euclidean algorithm:

$2703 = 1411 \cdot 1 + 1292$.

$1411 = 1292 \cdot 1 + 119$.

$1292 = 119 \cdot 10 + 102$.

$119 = 102 \cdot 1 + 17$.

$102 = 17 \cdot 6 + 0$.

$\Rightarrow \gcd(1411, 2703) = 17$ ✓

ii, Convert: 8 = 1000, F = 1111, B = 1011.

Thus, $(8FB)_{16} = (1000\ 1111\ 1011)_2$

Therefore, $(8FB)_{16} = (100011111011)_2$ ✓

iii, $(1001\ 011\ 0101)_2 = 010\ 010\ 110\ 101$.

Convert: 010 = 2, 010 = 2, 110 = 6, 101 = 5.

Hence: $(1001\ 011\ 0101)_2 = (2265)_8$.

Therefore: $(100\ 101\ 101\ 01)_2 = (2265)_8$ ✓

8. (30 points) Show the steps of using modular exponentiation algorithm to find $13^{61} \bmod 65$

Step 1: Convert the exponent to binary.

- The exponent is 61. We convert it to binary by finding the powers of 2: $61 = 32 + 16 + 8 + 4 + 1$.

$61 = (111101)_2$.

Step 2: Calculate the square of the base modulo 65.

- We compute $13^{2^k} \pmod{65}$ for each bit position:

+ $13^1 \equiv 13 \pmod{65}$.
+ $13^2 = 169 \equiv 39 \pmod{65}$.
+ $13^4 = 39^2 = 1521 \equiv 26 \pmod{65}$.
+ $13^8 = 26^2 = 676 \equiv 26 \pmod{65}$.
+ $13^{16} = 26^2 \equiv 26 \pmod{65}$.
+ $13^{32} = 26^2 \equiv 26 \pmod{65}$.

Step 3: Multiply the results according to the binary expansion.

- We multiply the values corresponding to the "1" bits in the binary representation $(111101)_2$:

$$13^{61} = 13^{32} \cdot 13^{16} \cdot 13^8 \cdot 13^4 \cdot 13^1 \pmod{65}.$$

Substitute in the values calculated in step 2:

1, $13^{32} \cdot 13^{16} = 26 \cdot 26 \equiv 26 \pmod{65}$.
2, $26 \cdot 13^8 \equiv 26 \cdot 26 \equiv 26 \pmod{65}$.
3, $26 \cdot 13^4 \equiv 26 \cdot 26 \equiv 26 \pmod{65}$.
4, $26 \cdot 13^1 \equiv 26 \cdot 13 = 338 \pmod{65}$.

Step 4: Final calculation.

$338 / 65 = 5$ remainder 13.

$338 = 5 \cdot 65 + 13$.

Answer: $13^{61} \equiv 13 \pmod{65}$.

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